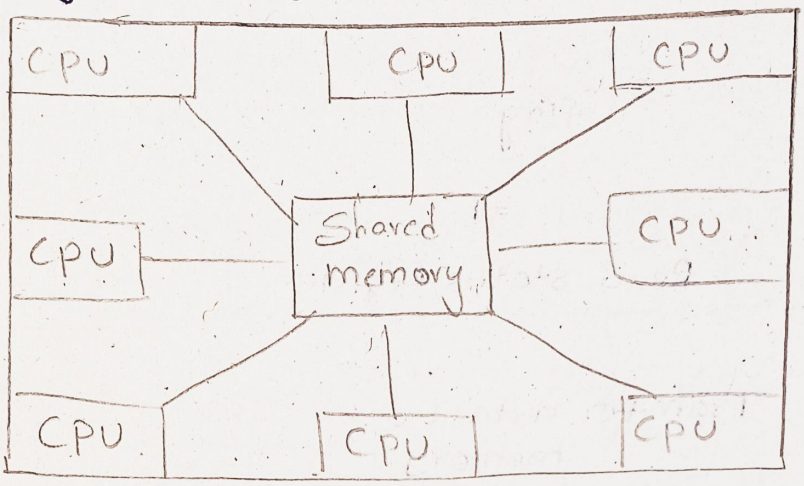
Unit-5

Multi Processors

*Multi processors :- A computer system with two or more CPU's sharing common memory & I/O devices for parallel processing and high speed execution.



Characteristics of multi processors :-

1. Reliability.
2. scalability.
3. Improved performance
4. Multiple applications.

① Reliability :-

→ failure of one CPU does not effect the system performance.

→ The work is distributed among diff CPUs

② scalability :-

→ The no. of CPU's can be increased based on user.

③ Improved performance :-

→ Multiple CPU's are used for executing diff types instructions in the same time. as a result system performance increases.

④ Multiple applications :-

→ It can be used as SISD (single instruction single data) and SIMD (single instruction multiple data) and MISD (multiple instructions single data) and MIMD (multiple instruction single data).

* Types of Multiprocessors :-

→ Implemented in two methods.

* shared memory technique → common memory for all the cpu's is used.

* Distributed memory technique → each CPU uses its own private memory.

** Inter connection of Multiprocessors :-

→ It gives the connection b/w cpu & memory modules.

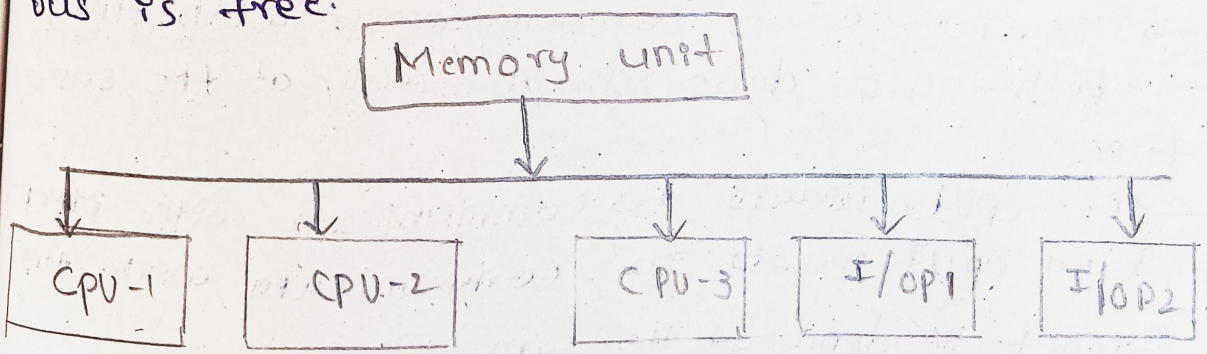
→ Implemented by 3 methods.

1. Time shared common bus.
2. Multiport memories.
3. Crossbar switch network.

① Time shared common bus :-

* In this common memory is shared among diff cpu using single bus for any communication. single bus is only responsible.

* If CPU1 wants to communicate with memory remaining cpu's should in waiting state until bus is free.



Advantages

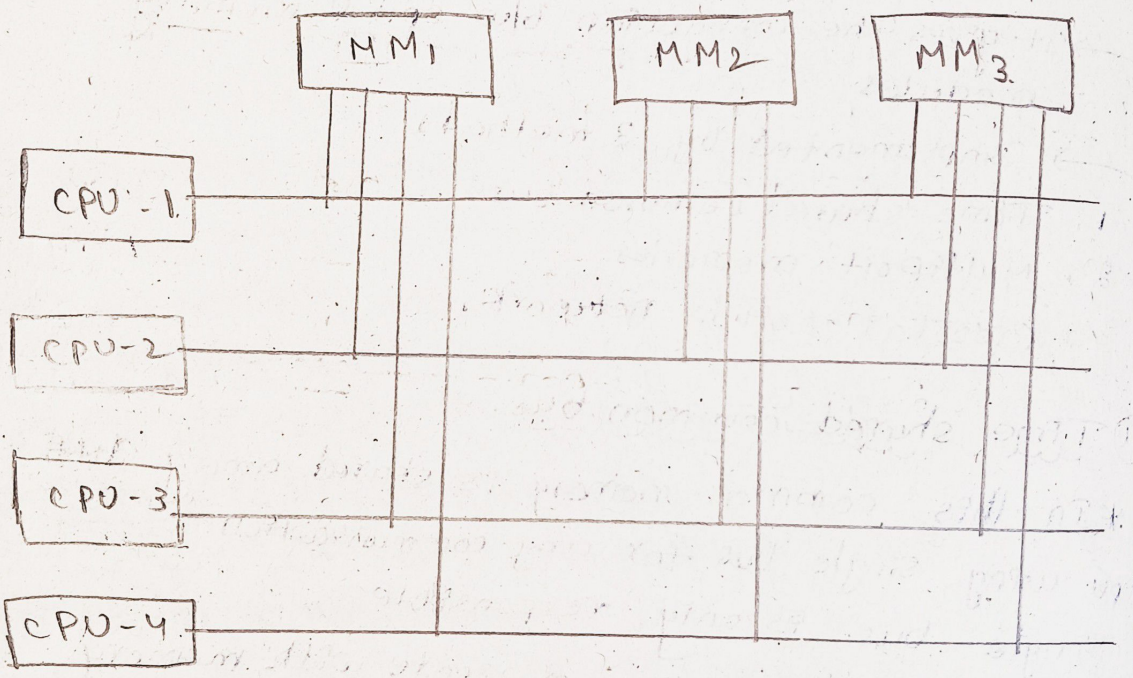
* Simple to implement
* low cost & complexity

Disadvantages

* Data transfer is low
* CPU's has to be in waiting state.

② Multiport memories :-

* In this memory is divided into diff modules along with CPU's
* Diff busses are used for connecting CPU's and every memory module.



Explanation :-

→ If CPU1 wants to communicate with MM1
→ If CPU3 wants to communicate with MM3
→ Both can be done simultaneously at the same time
→ If CPU1 wants to communicate with MM1
→ If CPU3 wants to communicate with MM1
Cannot be done at the same time.

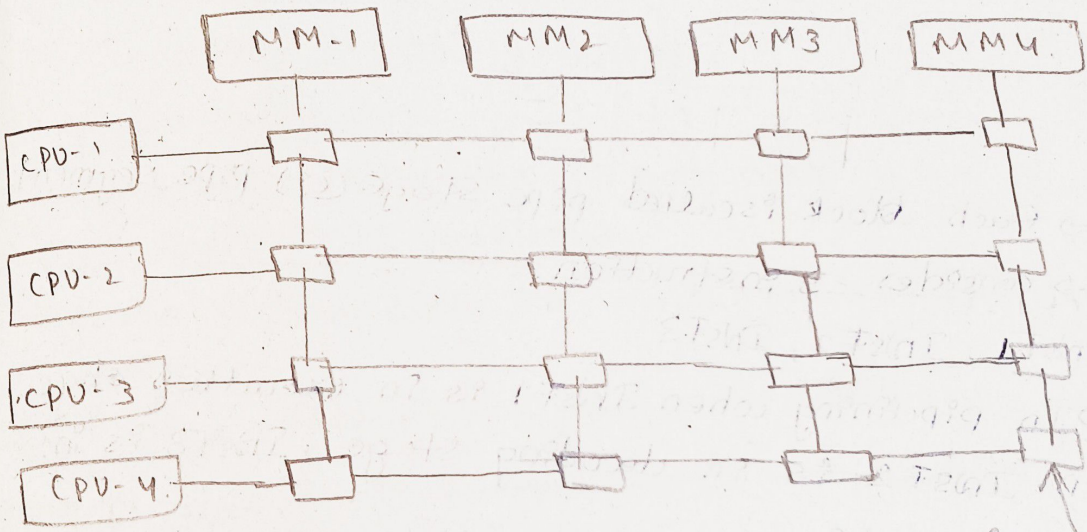
→ If that condition occurs priority comes into existance
→ High priority CPU communicates with memory 1st.

Advantage:-
* high speed data transfer.

disadvantage:-
* circuit complexity, high coast

③ Cross bar switch network:- → add matter as from multiport memories

* In this multiple busses are replaced with cross bar switch
* switch is responsible for passing the request b/w CPU and memory units



Advantage:- High speed data tranfer
disadvantage:- circuit complexity, high cost.

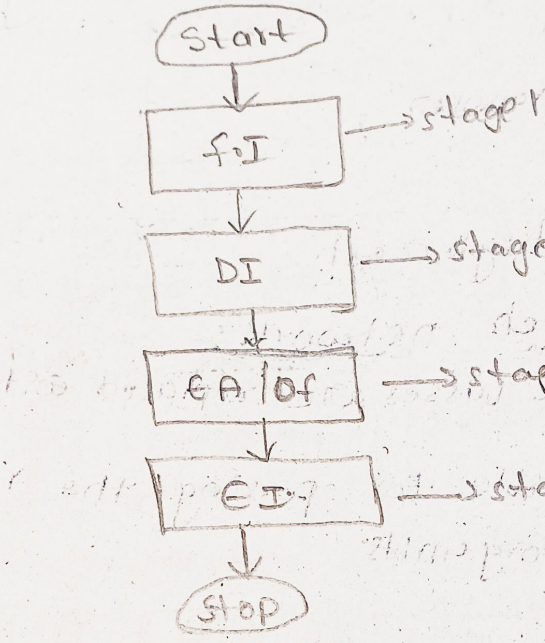
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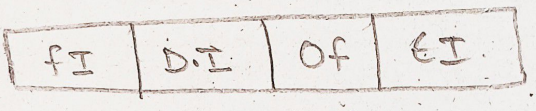
* Instruction ppelining:-
→ used to build fast processors.
→ It is a technique used to execute multiple instructions at the same time

* Instruction cycle:-
→ It is a part of instruction ppelining.

Stage-1 :- fetch the instruction
Stage-2 :- Decode the instruction.
Stage-3 :- find the effective address / operand fetch

step-4 :- execute the instruction.



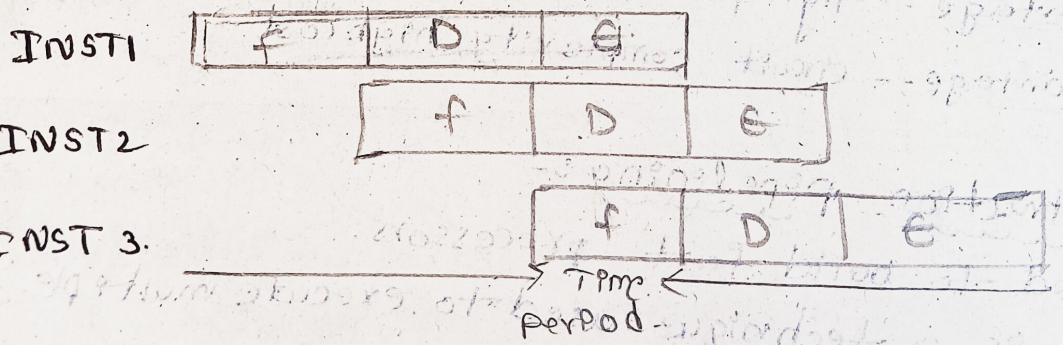


⟶ Each block is called pipe stage (or) pipe segment.

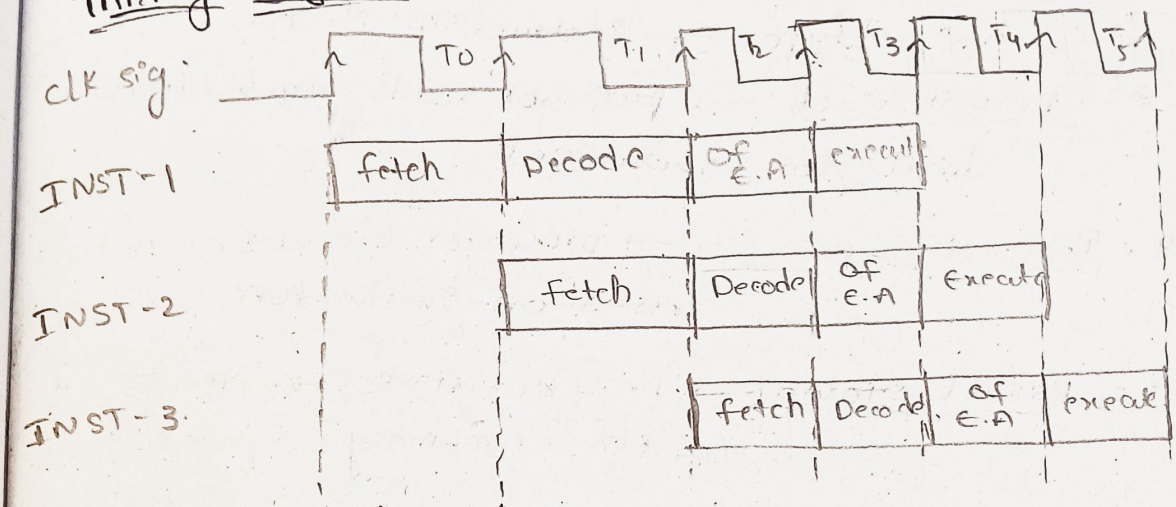
⟶ consider 3 instruction.

INST1, INST2, INST3

⟶ In pipelining when INST1 is in execution stage, then INST2 is in decoding stage, INST3 is in fetch state.



Timing diagram :-



Disadvantage of pipelining

1. Resource conflicts.
2. Data dependence conflicts.
3. Branching difficulty conflicts.

① Resource conflicts :-

→ occurs if initialization is not done properly

② Data dependence conflict :-

→ occurs when data is initialized at last for operation.

③ Branching difficulty conflict :-

→ occurs when iterative loops are used more in the program.

RISC pipelining :-

* features of RISC :-

→ simple instructions.
→ executes in one clock cycle.
→ fixed length instructions.
→ few addressing modes
→ High performance

Steages of pipelining

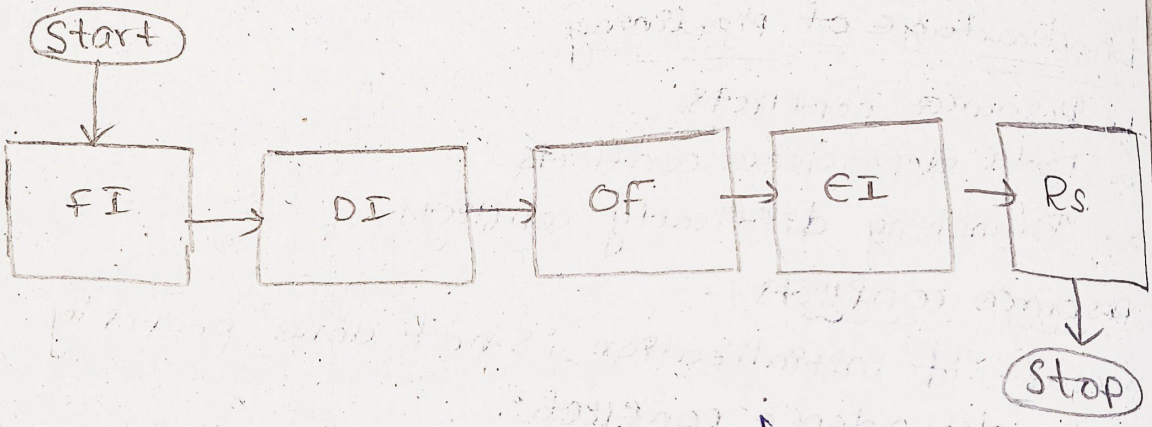
1. instruction fetch :- processor needs next instruction should be executed.

2. Instruction Decode:→ processor works on what type of instruction it is.

3. Operand fetch:→ processor reads req values of data for operations

4. Instruction execution→ processor executes on data based on instruction.

5. Result storage→ processor stores the result in register or memory.



→ RISC pipelining is also called as Assembly in line pipelining

* Parallel processing:-

→ "The principle in which more than one process works simultaneously to execute a program."

→ It is a part of multiprocessors

→ program instructions are divided among multiple CPU's for faster execution.

Advantages:-

→ uses parallel communication

→ High speed execution

→ saves execution time

* Types of parallel processing:-

Parallel processing



① SISD → single instruction single data.
→ follows sequential manner
→ executes one task at a time
→ found in single CPU systems.

② SIMD → single instruction multiple data
→ All instructions are taken by one CPU
→ these instructions are divided among multiple CPU's for execution.

③ MISD → Multiple instructions single Data.
→ Different instructions are executed at the same time.
→ follows the concept of pipelining

④ MIMD → Multiple instructions Multiple Data.
→ In this system has multiple CPU's where multiple tasks are performed at the same time.

* **Speed** :-
MIMD > MISD > SIMD > SISD.

* **cost** :-
MIMD > MISD > SIMD > SISD.

* **performance** :-
MIMD > MISD > SIMD > SISD

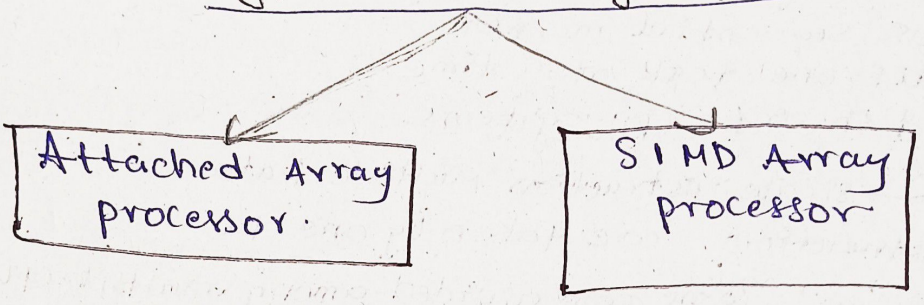
* **complexity** :-
MIMD > MISD > SIMD > SISD.

* **Array Processor** :-
→ processors are responsible for storing the data performing the required operations
→ Array processor follows specific architecture

for dealing with matrices and vectors.

→ Array processor can be self contained unit (or) distributed unit. Connected to memory and I/O devices using busses.

* types of Array processors



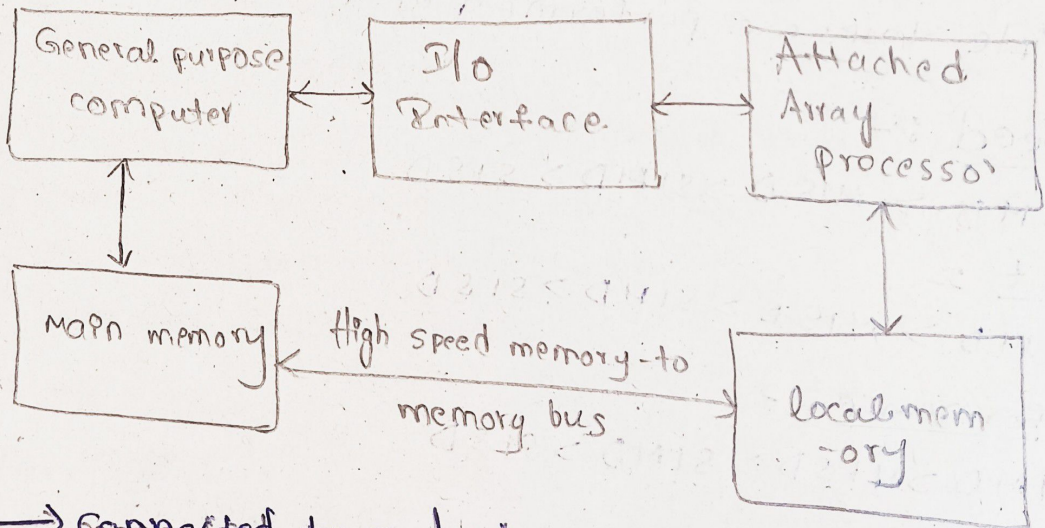
① Attached Array processor:-

→ works along with a main computer.

→ Main CPU gives instructions.

→ Array processor performs heavy calculations.

→ Results are sent back to the CPU



→ Connected to a host computer.

→ Used for numerical calculations.

→ Increases processing speed.

→ Works as a supporting processor.

→ It is used for scientific calculations.

→ also used in weather forecasting.

Advantages

→ Reduces load on main CPU.
→ faster Mathematical computations.
→ Efficient for scientific applications.

disadvantages

→ Expensive.
→ Requires coordination with main CPU.

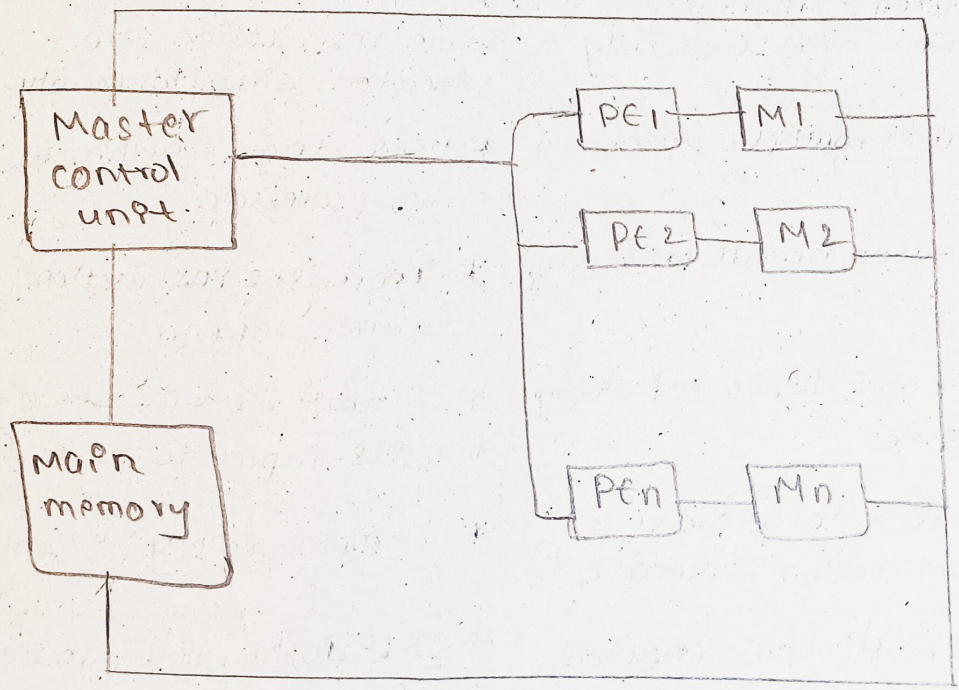
② SIMD Array processor :-

→ SIMD means :- Single Instruction multiple data.

* One instruction is applied to many data elements simultaneously.

→ A control unit sends the same instruction to all processing elements.

→ Each processing element works on different data at the same time.



→ It has parallel execution.
→ Same instruction for multiple data.
→ High speed computation.
→ Multiple processing elements.

Advantages:-

→ Very fast
→ Efficient for array operations.
→ Better performance in image processing

disadvantage:-

→ complex Hardware.
→ Not suitable for sequential tasks.

Difference b/w parallel processing and pipelining	
Parallel processing	pipelining
* Multiple tasks are executed at the same time using more than one processor or unit.	* A task is divided into stages, and diff stages work simultaneously.
* Many processors work together	* One process is split into steps
* Different instructions are executed simultaneously	* Different parts of the same instruction are executed simultaneously.
* Requires multiple processing units.	* uses stages within a single processor.
* Increases overall processing power.	* Increases instruction execution speed.
* Tasks are distributed among processors.	* Output of one stage becomes input to next stage.
* More complex because synchronization is needed.	* comparatively simpler.
* Ex:- Multi-core computers, super computers.	* Ex:- Instruction pipeline in CPU.
* can handle many tasks together	* Reduces execution time of instructions.
* Expensive hardware required	* Delay may occur due to pipeline Hazards.